Customer Shopping Behaviour Analysis

Uncovering insights from 3,900 purchases to drive strategic business decisions

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Dataset Overview

3,900

Transactions

Purchase records analyzed

18

Features

Data points per transaction

37

Missing Values

In review ratings column

Customer Data

- Age, Gender, Location
- Subscription Status
- Previous Purchases

Purchase Details

- Item, Category, Amount
- Season, Size, Color
- Discount & Promo Codes

Behavior Metrics

- Frequency & Timing
- Review Ratings
- Shipping Type

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Data Preparation Process

Data Loading

Imported dataset using pandas

Exploration

Checked structure and summary statistics

Cleaning

Handled missing values, standardized columns

Integration

Loaded to PostgreSQL for analysis

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Feature Engineering Highlights

Age Groups

Binned customer ages into meaningful segments

Purchase Frequency

Calculated days between purchases

Rating Imputation

Median-based imputation by product category

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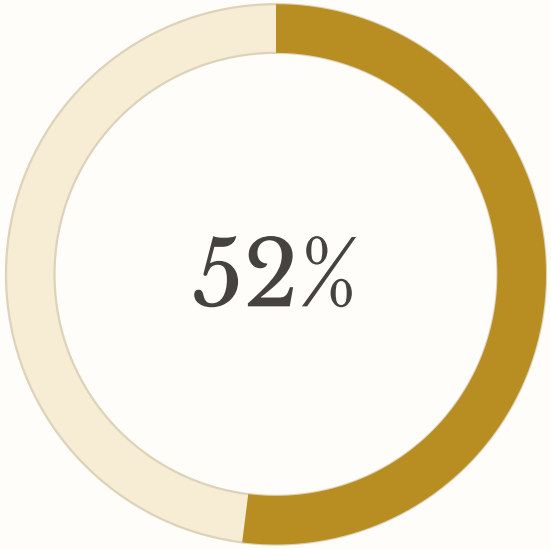
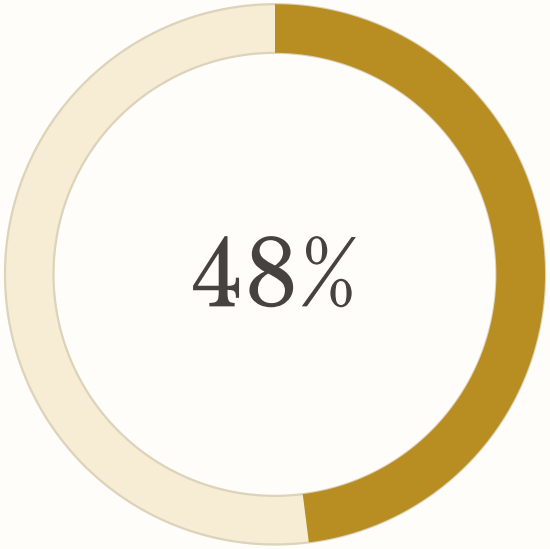
Revenue by Gender



Female

Share of total revenue



Male

Share of total revenue

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High-Spending Discount Users

Key Finding

Customers using discounts still spent above average purchase amount

Strategic Insight

Discounts drive volume without sacrificing per-transaction value

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Top Products by Rating

Leather Handbag

Average rating: 4.8 stars

Wireless Headphones

Average rating: 4.7 stars

Smartwatch

Average rating: 4.6 stars

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Shipping & Subscription Insights

Express Shipping

Higher average purchase amounts vs. standard shipping

Subscribers

Higher average spend and total revenue vs. non-subscribers

Repeat Buyers

Customers with >5 purchases more likely to subscribe

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Customer Segmentation

Loyal

Multiple purchases, high engagement

Returning

Second purchase, building relationship

New

First-time customers

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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers

Review Discount Policy

Balance sales boosts with margin control

Product Positioning

Highlight top-rated and best-selling products

Loyalty Programs

Reward repeat buyers to move into loyal segment

Targeted Marketing

Focus on high-revenue age groups and express-shipping users

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